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Molecular identification and antimicrobial resistance patterns of enterobacterales in community urinary tract infections among indigenous women in Ecuador: addressing microbiological misidentification

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Abstract

Background Antibiotic resistance of Enterobacterales poses a major challenge in the treatment of urinary tract infections (UTIs). In low- and middle-income countries (LMICs), standard microbiological (i.e. urine culture and simple disk diffusion test) methods are considered the "gold standard" for bacterial identification and drug susceptibility testing, while PCR and DNA sequencing are less commonly used. In this study, we aimed to re-identifying Enterobacterales as the primary bacterial agents responsible for urinary tract infections (UTIs) by comparing the sensitivity and specificity of traditional microbiological methods with advanced molecular techniques for the detection of uropathogens in indigenous women from Otavalo, Ecuador.

Methods A facility-based cross-sectional study was conducted from October 2021 to February 2022 among Kichwa-Otavalo women. Pathogens from urine samples were identified using culture and biochemical typing. Morphological identification was doble-checked through PCR and DNA sequencing of 16S, *recA*, and *rpoB* molecular barcodes. The isolates were subjected to antimicrobial susceptibility-testing using disk diffusion test.

Results This study highlighted a 32% misidentification rate between biochemical and molecular identification. Using traditional methods, *E. coli* was 26.19% underrepresented meanwhile *Klebsiella oxytoca* was overrepresented by 92.86%. Furthermore, the genera *Pseudomonas*, *Proteus,* and *Serratia* were confirmed to be *E. coli* and *Klebsiella spp.* by molecular method, and one *Klebsiella spp.* was reidentified as *Enterobacter* spp. The susceptibility profile showed that 59% of the isolates were multidrug resistant strains and 31% produced extended spectrum beta-lactamases (ESBLs). Co-trimoxazole was the least effective antibiotic with 61% of the isolates resistant. Compared to previous reports, resistance to nitrofurantoin and fosfomycin showed an increase in resistance by 25% and 15%, respectively.

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Conclusions Community-acquired UTIs in indigenous women in Otavalo were primarily caused by *E. coli* and *Klebsiella* spp. Molecular identification (16S/rpoB/recA) revealed a high rate of misidentification by standard biochemical and microbiological techniques, which could lead to incorrect antibiotic prescriptions. UTI isolates in this population displayed higher levels of resistance to commonly used antibiotics compared with non-indigenous groups. Accurate identification of pathogens causing UTIs and their antibiotic susceptibility in local populations is important for local antibiotic prescribing guidelines.

Keywords UTI, Indigenous women, *Shigella spp.*, *E. coli*, RpoB, RecA, 16S rRNA

Background

In patients with urinary tract infections (UTIs) it is critical to accurately identify the causal pathogen(s) and its antimicrobial resistance pattern. Uncomplicated UTIs is one of the most common indications for empirical antibiotic use in the community, that should be guided by local resistance patterns [1] UTIs caused by Gram-negative bacteria of the order Enterobacterales are by far the most prevalent in populations of women in low- and middle-income countries (LMICs) [2]. The bacteria causing UTIs and their antibiotic resistance patterns can differ depending on whether the infection is community-acquired (CA-UTI) or hospital-acquired (HA-UTI).

In CA-UTIs the uropathogens commonly include *Escherichia coli* (50–80% of cases), followed by *Klebsiella pneumoniae*, *Proteus mirabilis*, *Staphylococcus saprophyticus* and *Enterococcus* species [3, 4]. *Shigella* spp. have also been identified, albeit less frequently [5, 6]. These bacteria typically exhibit resistance to commonly prescribed antibiotics, such as ampicillin, co-trimoxazole, and fluoroquinolones (e.g., ciprofloxacin). Generally, antibiotics like nitrofurantoin and fosfomycin show lower resistance rates in CA-UTI pathogens. In HA-UTIs the causative bacteria are more diverse and often are more antibiotic resistant, *E. coli* remains a common player, but other species like *Pseudomonas aeruginosa*, *Candida* species, and *Enterobacter* spp. have been found to be more prominent and antibiotic-resistant [1, 7].

Globally, over 20% of *E. coli* isolates – the most common pathogen in UTIs – were resistant to both first-line antibiotics (ampicillin and co-trimoxazole) and second-line treatments (fluoroquinolones) [8]. The spread of ESBL-producing bacteria has made empirical treatment of UTIs more difficult, including resistance to common beta-lactam antibiotics [9]. Drug resistant uropathogenic *E. coli* has shown seasonality and associations with patients living in a rural setting; for example, resistance to ciprofloxacin, levofloxacin, and nitrofurantoin was higher among rural [10].

In Ecuador, antibiotic resistance rates in UTIs of pregnant women are high for ampicillin, ampicillin/sulbactam, amoxicillin, amoxicillin/clavulanic acid, and co-trimoxazole; therefore, empirical treatment with these drugs is not recommended; nitrofurantoin, fosfomycin or second-generation cephalosporins are empirically recommended [11]. Most UTI diagnoses conducted by the country's public health network are documented in the 'National Antimicrobial Resistance System' [12]. Ecuador has an estimated 66,000 UTI cases a year. According to data obtained from INEC (Ecuadorian National Institute of Statistics and Census), UTIs are the ninth cause of female morbidity and mortality in the country, with 10,522 hospital discharges [13]. Not all laboratories were able to identify the causative bacteria, particularly in low-income rural areas of the country, where the lack of budget, reagents, and health infrastructure is pronounced [14]. Besides, no study has been performed on the indigenous women of the Kichwa-Otavalo ethnic group living in the central Andes highlands. A report from 2014 by the Local Government (GAD) of Otavalo, found that UTIs are the second highest illness treated in the public hospital. It is therefore important to determine the pathogens and their antibiotic resistance causing UTI in this ethnic group [15].

Urine culture is considered the "gold standard" in the identification of bacteria causing UTIs. It is time-consuming and in approximately 20% of patients with UTI symptoms produces false-negative results. Antimicrobial susceptibility of isolated bacteria is typically assessed using the simple diffusion disk method (Kirby-Bauer). Results are generally available 48 to 72 h after sampling [16]. To ensure timely initiation of treatment, even in the absence of identified causative bacteria, empirical antibiotics such as nitrofurantoin, fosfomycin, or second-generation cephalosporins, are recommended [11]. Globally, DNA-based methods, such as polymerase chain reaction (PCR) and DNA sequencing, are increasingly being adopted and have the potential to replace traditional microbiological techniques for the accurate identification of pathogens [17]. 16S ribosomal RNA (rRNA) analysis has been widely used for the identification of uropathogens [18]. The bacterial 16S rRNA gene contains highly conserved and hypervariable regions between heterogeneous and homogeneous species, respectively [19]. 16S rRNA gene sequencing is highly sensitive for the identification of pathogens from clinical samples [20]. However,

it is unable to differentiate *Shigella* spp. from *E. coli* [21]. Due to this low specificity, it is necessary to use additional molecular markers, such as genes encoding the subunit of bacterial recombinase (*recA*) and the beta-subunit of RNA polymerase (*rpoB*), that proved to be valuable for bacterial species assignment. [22–25]. These three genes together can provide better and faster discrimination of Enterobacterales species causing UTIs.

Although UTIs are a significant cause of morbidity in Ecuadorian women, there is little to no information in more rural settings on uropathogens and their antibiotic susceptibility. Thus, this study aimed to identify the bacterial agents responsible for UTIs among Indigenous women from a rural town in Ecuador, to compare uropathogenic detection using microbiological techniques versus molecular methods; and to evaluate antibiotic susceptibility patterns following Clinical and Laboratory Standards Institute (CLSI) guidelines.

Materials and methods

Study area and sampling

A cross-sectional study was conducted from October 2021 to February 2022, in which a total of 381 urine samples were collected. The outpatients were women over 18 years old who had voluntarily presented themselves at the laboratories for diagnostic testing in Otavalo, Ecuador (Figure S2, Table S4). Neither hospitalized patients nor those with a history of hospitalization were included in the study. Additionally, women who had received antibiotics within the preceding 15 days were excluded from the analysis. Urine samples were collected in sterile containers at the laboratory during the morning, following standard procedures outlined in the Ecuadorian Ministry of Health Manual [11]. After sterilization, the remaining samples, which did not display any evidence of infection, were discarded as biological waste per the protocols established by the laboratory and the municipal regulations of Otavalo.

Bacterial isolation

From the 381 urine samples obtained from outpatients, only those that exhibited signs of infection upon microscopic examination were selected (presence of bacteria and white blood cells. Once potential UTI samples were identified, bacterial isolation was carried out with 10 µL of sample on MacConkey agar medium (Becton Dickinson, New Jersey, USA), for the isolation of Enterobacterales and subsequent counting. Typical colonies, distinguished by their purple and creamy white appearance as well as lactose-negative and lactose-positive characteristics, were identified and quantified. Microorganism counts were expressed as colony-forming units per milliliter (CFU/mL) of urine. A positive diagnosis for urinary tract infection was assigned to all samples where the microorganism count exceeded 10^5 CFU/mL.

Phenotypic identification

A colony was selected from each infection-positive sample for biochemical tests to identify the species of bacteria. Suspected colonies were isolated and further confirmed as Enterobacterales through a series of biochemical tests. These tests included inoculation onto Simmons citrate medium, triple sugar iron, urea, sulfur indole motility, malonate, and the methyl red/Voges–Proskauer tests (all media and solutions were from Becton Dickinson, USA). All isolates were preserved in brain heart infusion broth with 10% glycerol. Those used in the study were frozen at − 20 °C, and those preserved for future experiments were frozen at − 80 °C (ATCC, Manassas, Virginia, USA).

Antibiotic susceptibility test

The Kirby Bauer disk diffusion method was performed following the guidelines of the Clinical & Laboratory Standards Institute (CLSI) [26]. The antibiotics used were as follows: amikacin (30 µg); amoxicillin/clavulanic acid (30 µg); gentamicin (30 µg); ceftriaxone (30 µg); cefuroxime (30 µg); nitrofurantoin (300 µg); fosfomycin (50 µg); levofloxacin (5 µg); nalidixic acid (30 µg); ampicillin/sulbactam (30 µg); co-trimoxazole (30 µg). Additionally, the double-disk synergy test (DDST) was performed to obtain the phenotypic identification of Extended-Spectrum Beta-Lactamase (ESBL) in Enterobacterales, using individual Petri dishes with the recommended antibiotics: Amoxicillin/clavulanic acid (30 µg); Ceftriaxone (30 µg); Ceftazidime (30 µg); Cefepime (30 µg); Aztreonam (30 µg) (all antibiotic discs were from Thermo Scientific™ Oxoid™). The strain *E. coli* ATCC 25922 was used as a control.

Genomic DNA extraction and quantification

For DNA extraction, a single colony from each pure culture was taken and resuspended in TE buffer. DNA extraction was performed using the Chelex-10 method described by Suenaga & Nakamura [27], followed by proteinase K digestion. The supernatants were recovered in microtubes and stored at -20 °C. DNA samples were analyzed using a NanoDrop instrument (Thermo Scientific™, Waltham, MA, USA, ND-2000) to determine DNA quantity and purity. The samples were then stored at –20 °C for further analysis.

Molecular identification (DNA amplification and sequencing)

The housekeeping genes 16S rRNA, *recA*, and *rpoB* were used as markers for the molecular identification of

isolated strains, by PCR. The 16S rRNA V1-V3 region was amplified using primers 8F (5'-AGAGTTTGATCC TGGCTCAG-3') and 534R (5'-ATTACCGCGGCT GCTGG-3'), following the protocol of Chen et al. [28]. A region of 1074 bp of the beta-subunit of bacterial RNA polymerase was amplified with primers rpoBF (5'- GGC GAAATGGCWGAGAACCA-3') and rpoBR (5'- GAG TCTTCGAAGTTGTAACC-3') [29]. The PCR conditions were as follows: initial denaturation at 95 °C for 5 min, 35 cycles of denaturation at 95 °C for 30 s, annealing at 50 °C for 30 s, extension at 72 °C for 30 s, and a final extension at 72 °C for 5 min. For *recA* identification, a region of 780 bp was amplified with recAF (5'-CGCATTCGC TTTACCCTGACC-3') and recAR (5'-TCGTCGAAA TCTACGGACCGGA-3') [30] following the previously described conditions with an annealing temperature of 58 °C. Amplification was performed in reactions of 15 µL containing 2X GoTaq® Green Master Mix (Promega), 0.3 µM of each primer, and 0.33 to 1.7 ng/µL DNA.

PCR products were sequenced using the Sanger technique on an ABI 3500xL Genetic Analyzer (Applied Biosystems, Foster City, CA, USA) with BigDye 3.1® capillary electrophoresis matrix. Sequences were edited using MEGA X software [31] and compared against the GenBank database at the National Center for Biotechnology Information (NCBI) through the Basic Local Alignment Search Tool (BLAST). Samples with overlapped wavelengths in the electropherogram were re-isolated for the assessment of mixed cultures.

Phylogenetic analysis

The sequences were aligned using the MAFT algorithm, and a best-fit model was selected based on the Akaike Information Criterion (AIC), utilizing the option for the best DNA/protein model. Tree topologies and branch lengths were computed using the maximum-likelihood method (ML) with 500 bootstrap replicates. The models employed were General Time Reversible (GTR) with a gamma distribution (G) and invariant sites (I) for 16S rRNA, Tamura-Nei 93 (TN93) + G + I for *rpoB* and *recA*, and GTR + I for the concatenated sequences. The iTOL software [32] was employed for tree visualization and annotation. The 16S rRNA/*rpoB*/*recA* concatenated construct was generated using the NGPhylogeny online software [33].

Results

Microbiological identification of UTIs isolates

A total of 136 isolates (S1-S136) were cultured at the clinical laboratories in Otavalo town. Using the standard microbiological methods, all isolates were identified as Enterobacterales comprising *Escherichia coli* (68.4%, 93), *Proteus mirabilis* (11%, 15), *Klebsiella oxytoca* (10.3%, 14), *Klebsiella pneumoniae* (5.9%, 8), *Pseudomonas* spp, (2.9%, 4), and *Serratia marcescens* (1.5%, 2).

Molecular identification of UTIs isolates

To verify morphological identification, we conducted several PCRs to partially amplify the three genes 16S rRNA, *recA*, and *rpoB* in the 136 isolates obtained. The sequencing of seven strains (S10, S44, S53, S54, S61, S66, and S67) exhibited double peaks in the electropherogram signal, leading us to suspect they might represent consortia. Therefore, we reisolated these strains until obtaining two clean 16S rRNA sequences for each one. Thus, resulting in seven additional strains, summing up 143 strains. We subsequently renamed each strain by appending "0.1" and "0.2" to their previous codes (Suppl. Table 4). Ultimately, we obtained 429 sequences, with three for each strain, except for S65 and S108 where *recA* did not amplify.

To conduct a comprehensive molecular analysis and facilitate molecular classification visualization, we grouped gene sequences based on their similarities. For this purpose, we compared each gene sequence with reference sequences to identify genetic variants (Suppl. Tables 1–3), considering at least one nucleotide change compared to the reference sequence. A total of 69 variants were identified (Table 1) and distributed among the three genes used: 24 for 16S rRNA (EC 16SrRNA-V1-10; KP 16SrRNA-V1-11; KO 16SrRNA-V1; KA 16SrRNA-V1; EH 16SrRNA-V1), 21 for *rpoB* (EC rpoB-V1-12; KP rpoB-V1-6; KO rpoB-V1; KA rpoB-V1; EH rpoB-V1) and 24 for *recA* (EC recA-V1-17; KP recA-V1-6; EH rpoB-V1). These sequences were cross-referenced with the GenBank database, resulting in the identification of five bacterial species: *E. coli* (88%, $n=126$), *K. pneumoniae* (10%, $n=14$), *K. oxytoca*, *K. aerogenes*, and *Enterobacter hormaechei* (0.7%, $n=1$).

Regarding the 16S rRNA gene, sequences matched 100% both *E. coli* and *Shigella flexnerii* or *E. coli* and *Shigella sonnei* in 94.4% of the strains (70% of variants). Only the variants EC 16SrRNA-V2/V5/V8 were specific to *E. coli*. In terms of the *rpoB* gene, 47.6% of *E. coli* variants matched 100% also with *Shigella* species such as *S. flexneri* (EC rpoB-V4), *S. dysenteriae*, and *S. flexneri* (EC rpoB-V1/V11), as well as *S. boydii* (EC rpoB-V9). Additionally, with the *recA* gene, 22.2% matched 100% *E. coli* and *S. flexneri* (EC recA-V6) or *E. coli* and *S. sonnei* (EC recA-V9) (refer to Supp. Table 4). The *recA* gene exhibited the highest potential for identifying *E. coli* at 69% ($n=98$), followed by *rpoB* at 46% ($n=66$).

Comparing conventional against molecular identification methods, we observed that out of the 14 isolates microbiologically identified as *K. oxytoca*, 10 were re-identified as *E. coli,* 2 were re-identified as *K.*

Table 1 Molecular variants and reference sequences for 16S rRNA, *rpoB*, and *recA* molecular markers

Bacterial spp.	Molecular markers					
	16S rRNA		*rpoB*		*recA*	
	Ref. sequence	Variants (V)	Ref. sequence	Variants (V)	Ref. sequence	Variants (V)
Escherichia coli	AB681728.1	10	GCF_000005845.2	12	GCF_000005845.2	17
Klebsiella pneumoniae	NR_036794.1	11	GCF_002903005.1	6	GCF_002903005.1	6
Klebsiella oxytoca	AB681870.1	1	GCF_003812925.1	1	-	-
Klebsiella aerogenes	AJ251468.1	1	GCF_007632255.1	1	-	-
Enterobacter hormaechei	FP929040.1	1	GCF_019048625.1	1	GCF_019048625.1	1

pneumoniae, one as *Enterobacter hormaechei*, and one as *K. aerogenes*. Among the 15 *P. mirabilis*, 13 were re-identified as *E. coli* and 2 as *K. pneumoniae* (Supplementary Table 4). Only three strains of *K. pneumoniae* (38%) and 91 strains of *E. coli* (98%) isolates matched both conventional and molecular methods. On the other hand, *Pseudomonas* spp. was re-identified as *E. coli* by molecular methods, while the 2 *S. marcescens* were re-identified as *E. coli* and *K. oxytoca*. Finally, *K. aerogenes* and *E. hormaechei* were not identified morphologically, conversely *P. mirabilis* and *S. marcescens* were not detected through molecular tools (Figure S2).

However, the three barcodes did not differentiate between *E. coli* and *Shigella* species in 99% of the strains due to their high (or identical) similarity with the reference sequences in GenBank. Although the combination of 16S rRNA, *rpoB*, and *recA* genes allowed the confirmation of most of the bacterial species, it did not enable the distinction between *E. coli* and *Shigella*. To further differentiate between these two genera, a phylogenetic tree was constructed using the concatenated three genes (16S/*rpoB*/*recA*) with the reference sequences of *Shigella* species and *E. coli* (Fig. 1). The resulting tree grouped *S. sonnei* and *S. flexneri* in one cluster along with S29, S67, S101, S127, S128, and S123.

Antibiotic resistance profile

After molecular identification, the morphologically misidentified MDR strains turned out to be E. coli (S8, S10.2, S13-17, S19, S22, S31-33, S36, S39, S41, S44.1, S46, S53.1, S54.1, S57, S63, S67.1, S75-76, S95, S100, S122, S125-126, S128, S131-135), *Enterobacter hormaechei* (S44), *Klebsiella oxytoca* (S65), *Klebsiella pneumoniae* (S98, S103, S112-4), and *Klebsiella aerogenes* (S108) (Fig. 2). Overall, the prevalence of antibiotic resistance was as follows: Cotrimoxazole 61% ($n=87$); nalidixic acid and ampicillin/sulbactam 49% ($n=70$); cefuroxime 33% ($n=47$); ceftriaxone 30% ($n=43$); ceftazidime 29% ($n=42$); levofloxacin 27% ($n=38$); nitrofurantoin 25% ($n=36$); aztreonam 20% ($n=28$) and cefepime 17% ($n=25$) amikacin, amoxicillin/clavulanic acid and, fosfomycin 15% ($n=21$) and gentamicin 8% ($n=11$). Likewise, an important 31% ($n=44$) were phenotypically identified as ESBL isolates.

After molecular identification, the morphologically misidentified MDR strains turned out to be *E. coli*, except for S44 (*Enterobacter hormaechei*), S65 (*Klebsiella oxytoca*), S98, S103, S112-4 (*Klebsiella pneumoniae*), and S108 (*Klebsiella aerogenes*). The entire resistance profile and phylogenetic relationship of the strains are illustrated in Fig. 2.

Discussion

This study analyzed 136 isolates from community-acquired urinary tract infections (CA-UTIs) in Indigenous women, identifying all infecting bacteria as belonging to the Enterobacterales order. *E. coli* was the most common, detected by both microbiological (68%) and molecular (92%) methods. *E. coli* is known to ferment lactose, yet it has been reported that up to 10% of *E. coli* isolates are slow or non-lactose fermenters (NLF) due to a deficiency in the lactose permease enzyme. NLFs could be misidentified as other species since they present negative results in fermentation tests and form colorless colonies on selective media such as MacConkey agar [34]. Notably, non-lactose fermenting *E. coli* is more likely to be isolated from patients in the community [35]. This study is the first analyzing UTI isolates obtained from the indigenous Kichwa-Otavalo ethnic group residing in the rural Andes highlands of Ecuador. The crude prevalence of *E. coli* identified through conventional methods (68.4%) aligns with findings from previous studies conducted in the Andes of Ecuador on CA-UTIs using the same methodology. For instance, a study conducted in Quito reported a prevalence of 79.4% [36], while another study in Latacunga found a prevalence of 84.5% [37]. However, employing molecular methods in our study revealed a higher prevalence of 92%, surpassing the rates reported in a previous study on other indigenous ethnic

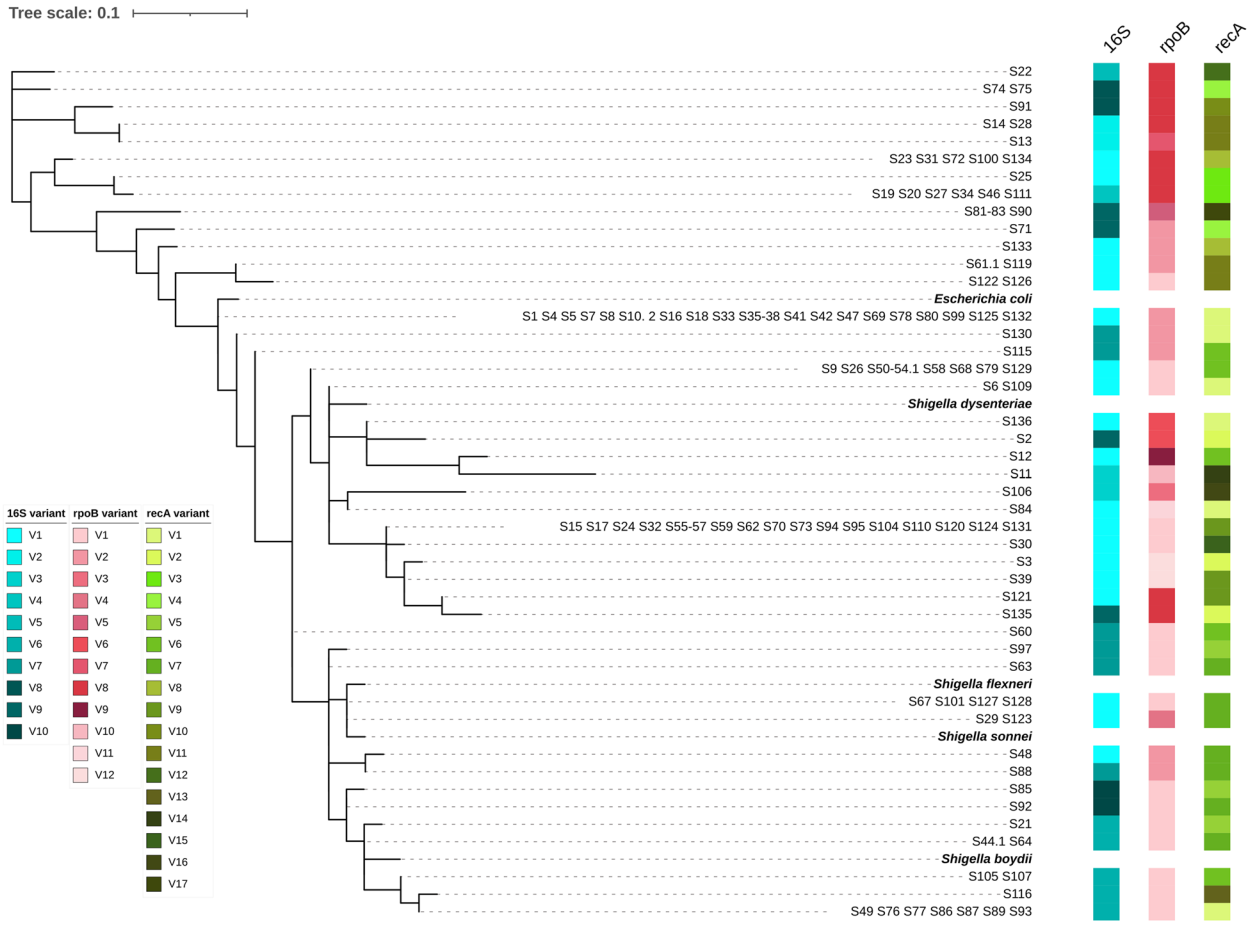


Fig. 1 Phylogenetic tree constructed with 16S rRNA/ *rpoB*/ *recA* concatenated sequences from strains with ambiguous identification (*E. coli* vs *Shigella* spp.) and reference sequences from GenBank. The phylogenetic tree was constructed using the maximum-likelihood method. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the General Time Reversible + G + I method

groups in the Andes, where *E. coli* prevalence was 83% [38]. Clearly, molecular characterization enhances the detection rate of *E. coli*. This finding supports previous studies suggesting that Enterobacterales species such a *Shigella* and *E. coli* should be classified as subgroups within the same species, as they are genetically very similar [39, 40].

This study represents the first attempt to compare microbiological and molecular methods for bacteria identification in CA-UTIs of indigenous Kichwa-Otavalo women. Although urine culture remains the 'gold standard' for UTI diagnosis and bacterial identification, infrastructure and reagents are still lacking in many low-resource settings within LMICs, such as Ecuador [14]. Although UTI-specific chromogenic media are recommended by protocols, they were not available in the studied laboratories. Instead, MacConkey agar medium was used, which presents several drawbacks. One of the main challenges is discriminating between fermenters and non-fermenters, as varying expression levels during fermentation can occur both within and between Enterobacterales species [41, 42]. This issue may introduce bias in the microbiological identification of UTI pathogens. Furthermore, it also requires the use of a specific battery of biochemical tests [43]. Altogether, it has the potential to induce UTI misdiagnosis. For this reason, molecular complementary techniques are recommended as is used in high-income countries [18]. After PCR analysis, it was revealed that up to 30% of isolates were misidentified by the microbiological methods. The high rate of discrepancies in the identification of pathogens in UTIs has underscored the need for the implementation of molecular tests as a means of confirming diagnoses in the study area, and potentially in LMICs more broadly.

We used three housekeeping genes (16S rRNA, *rpoB*, and *recA*) for strain identifications. Nevertheless, taxonomy at the species level was a difficult task since 70% of 16S rRNA *E. coli* variants were also identified as *Shigella*

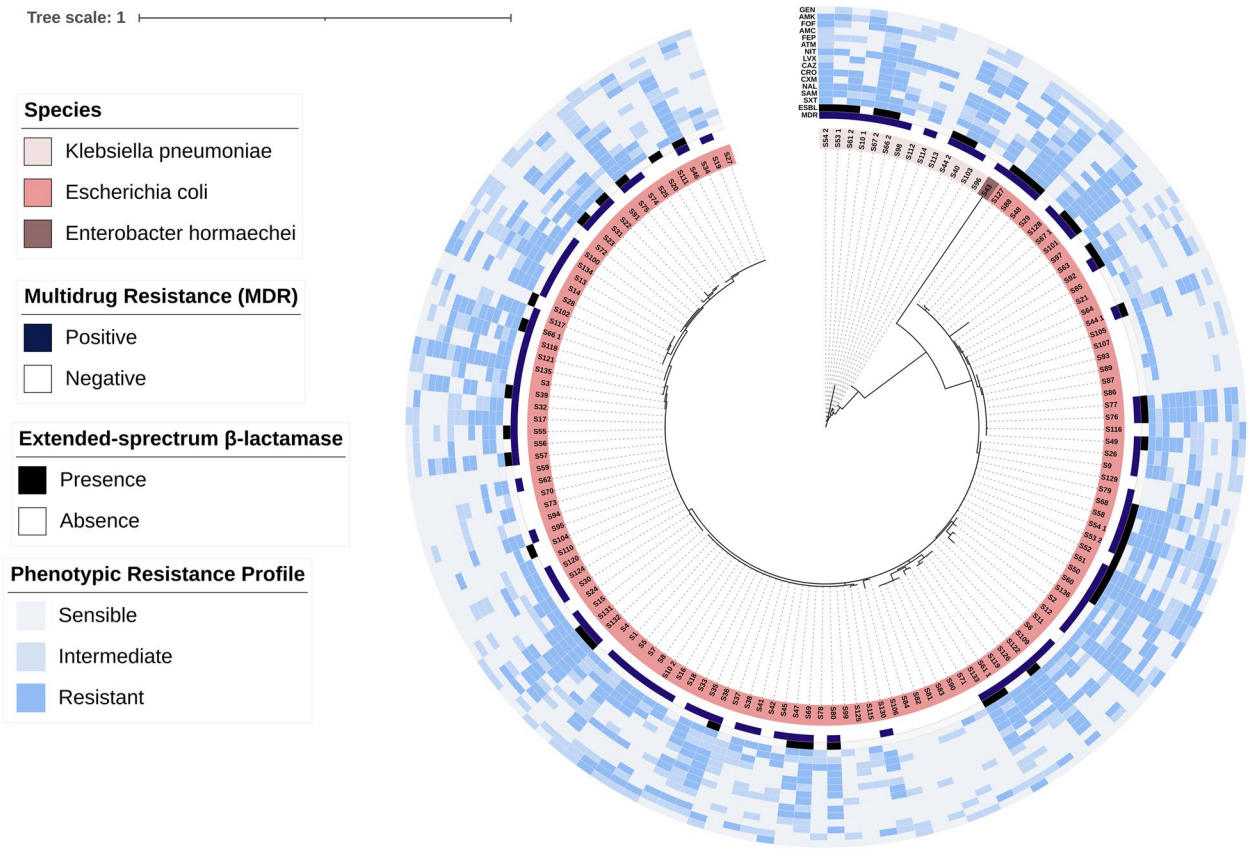


Fig. 2 Phylogenetic tree and antibiotic resistance profile of UTI strains of Ecuadorian indigenous women. It was constructed using the 16S rRNA/ *rpoB*/ *recA* concatenated sequences with the maximum-likelihood method. The isolates S65 and S108 were not included due to the lack of *recA* amplification. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the General Time Reversible + I method

spp., *S. flexneri*, and *S. sonnei* in the BLAST analysis. This is consistent with previous studies where low discriminatory power has been reported for 16S rRNA in closely related species, such as *Shigella* spp. and *E. coli* [44]. *rpoB* is known to have high resolution for differentiating species at a population level. However, it has been reported that *Shigella spp.* and *E. coli* share more than 93% similarities reaching up to 99.8% with S. *flexneri*, 99.4% with *S. sonnei*, and 99.78% with *S. boydii* [44] Nonetheless, Devanga Ragupathi et al. [45] revealed that *rpoB* accurately identified *Shigella spp.* and different *E. coli* strains. Thus, 47.6% of *E. coli* variants matched with 100% *S. dysenteriae*, *S. flexneri*, S. *boydii*, and *E. coli*. On the other hand, *recA* discriminated the most *E. coli* from *Shigella* spp., reducing species overlapping to 12%. Nevertheless, the three molecular barcodes combined were not able to differentiate 20% of the *E. coli* and *Shigella* spp. strains (Supp. Table 4). The phylogenetic tree from the combined genes grouped most of the strains with *E. coli* reference sequences. Nevertheless, strains S67, S101, S127, S128, and S29, S123 clustered with *S. flexneri* and *S. sonnei*, indicating they likely belong to the *Shigella* genus. Previous studies utilizing whole genome sequencing (WGS), SNP identification, and DNA-DNA hybridization have suggested that *Shigella* should be reclassified within *E. coli* due to their high genetic similarity [40, 41]; however, no changes have been implemented for medical purposes. UTIs caused by *Shigella* spp. are rare and are mainly associated with gastrointestinal infections, often through contaminated food and water [44, 46]. The only previous study in Indigenous Kichwa also identified *Shigella* spp.in 1% of UTIs [38]. Misidentification of the genus could lead to the selection of inappropriate empirical treatments, such as narrow-spectrum antibiotics that may not specifically target the genus in question, as observed with *Pseudomonas* spp. and *Klebsiella* spp. in this study. Such mistreatment can contribute to the development of antibiotic resistance while failing to address the underlying infection. Mistreatment can result in the development of antibiotic resistance without resolving any issues of underlying infection.

Important levels of antimicrobial resistance were detected in our study. We identified 85 Enterobacterales isolates as multidrug resistant (MDR), with 44 of these being ESBL producers. This poses a significant risk of complicated infections and difficult to cure with the conventional antibiotics used [47, 48] The spread of ESBL-producing bacteria has made empirical treatment more difficult and has promoted resistance to beta-lactam antibiotics, such as penicillin and cephalosporins [9]. Sixty-one percent of the isolates were resistant to co-trimoxazole, making it the least effective in our study. This resistance is comparable to previous rates described by Solís et al. [36] which showed 55% resistance to this antibiotic in CA-UTIs in the city of Quito. Another study from the same city on outpatients from a tertiary hospital reported 84% of bacteria being resistant to this antibiotic [49]. Additionally, nalidixic acid and ampicillin/sulbactam displayed high rates of resistance (49%). These results resemble those of studies of UTIs in urban areas of Ecuador, with prevalences of 68% [49] and 62% [36], respectively.

Interestingly, this study reveals concerning resistance rates for antibiotics that were previously reported as effective in treating uncomplicated UTIs [10]. For instance, we found 25% resistance to nitrofurantoin and 15% for fosfomycin, compared to previous Ecuadorian analyses. These resistance levels are increasing [36, 38]. For instance, resistance rates of 5% and 4% have been documented in isolates from hospitalized and ambulatory patients, respectively [36, 49]. Furthermore, the high rate of levofloxacin resistance found in this study (27%) should also be of concern. Quinolones/fluoroquinolones are reserved for complicated pyelonephritis or for patients unable to receive first-line antibiotics—nitrofurantoin, fosfomycin, or co-trimoxazole—for uncomplicated UTI [10, 11] At the community level, bacterial resistance has been linked to antibiotic misuse practices. The higher prevalence of antibiotic resistance found in this study may be due to factors that are present in low-income populations living in rural settings: 1) a greater frequency of self-medication; 2) less frequent bacterial identification and susceptibility testing compared to urban areas, which may act as a selective pressure mechanism; 3) antibiotics of every group are sold without medical prescription; 4) inadequate dosage; 5) little access of healthcare services; and 6) misuse and overuse of the recommended nitrofurantoin and fosfomycin [10, 14, 50]. This phenomenon is prevalent in rural Ecuador, attributed to the widespread practice of backyard animal husbandry and the unregulated use of antibiotics as growth promoters in animals [51–53]. Additionally, the existence of active horizontal gene transfer mechanisms, such as plasmid fusion, has been reported in human and animal isolates from Ecuador [54].

The present study has some limitations: 1) working with strains isolated in diagnostic laboratories, we lack knowledge of the individual resistance profiles of mixed strains that were subsequently separated, re-isolated, and characterized. Nevertheless, we ensure that the number of these mixed strains does not represent any bias in the final interpretation of our resistance rates; 2) the re-identification was limited to the use of three molecular markers. Nevertheless, the identification at species level was successful using the three specific molecular markers; 3) The sampling period of five months (October 2021 to February 2022) provides a meaningful snapshot of antimicrobial resistance patterns within the studied population. However, it is recognized that a longer sampling period could have offered a more comprehensive understanding of potential temporal variations, including seasonal fluctuations in resistance profiles.

Conclusions

There is still a long way to go to find solutions that can strengthen the traditional diagnostic procedures in LMICs. Uropathogenic bacteria identification using DNA analysis could be useful in the management of clinically suspected UTIs, strengthen antibiotic administration, and avoid complications and hospitalizations, decreasing the costs associated with the patient's management. The high bacterial misidentification results of our study imply the need for policymakers to incorporate DNA-based diagnostic laboratories located in rural towns into the national surveillance network.

Supplementary Information

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Supplementary Material 1.
Supplementary Material 2.
Supplementary Material 3.
Supplementary Material 4.
Supplementary Material 5.
Supplementary Material 6.

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Authors' contributions
CBC: experiment design, data analysis, writing, and editing. FHA: experiment design, laboratory experiments, data analysis, writing, and editing. MA: sample collection and laboratory experiments. MO: sample collection and laboratory experiments. JB: laboratory experiments and data analysis. JG: writing and editing. MC: writing and editing. PC: experiment design, data analysis, writing, and editing. All authors have read and agreed to the published version of the manuscript.

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Data availability
The sequences obtained were registered in the NCBI nucleotide database under the following accession number for 16S rRNA: PP236120-PP23643; recA: PP239204-PP239227 and rpoB: PP239228-PP239248. All data generated or analysed during this study are included in this published article and its Supplementary Fig. 1 and Supplementary Tables 1–4.

Declarations

Ethics approval and consent to participate
The statements on ethical approval and the informed consent were granted by the Human Research Ethics Committee of the Universidad de las Américas, Quito-Ecuador (code 2023EXC-014–2). All informed consent forms signed by patients are stored at CEMOPLAF facilities in Otavalo. CEMOPLAF's sampling and diagnostic permits were granted by the Ministry of Health of Ecuador under the code 34409.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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